



Logical & Set Operators

The symbols below use standard LaTeX & KaTeX math formatting (e.g., $\wedge$ for $\wedge$ ($\wedge$)), meaning you don't need to memorize difficult Alt codes or Hex inputs to type them!



QUICK CHEAT SHEET

LaTeX	Symbol	Concept	Meaning
$\neg$	$\neg$	NOT	Flip the truth value
$\wedge$	$\wedge$	AND or CONJUNCTION	Both must be true
$\vee$	$\vee$	OR or DISJUNCTION	At least one is true
$\oplus$	$\oplus$	XOR	Exactly one is true
$\rightarrow$	$\rightarrow$	IMPLIES or IF	If A, then B
$\leftrightarrow$	$\leftrightarrow$	BICONDITIONAL or IFF	Both match (if and only if)
$\forall$	$\forall$	FOR ALL	Everything satisfies
$\exists$	$\exists$	EXISTS	At least one satisfies
$\exists!$	$\exists!$	UNIQUE	Exactly one satisfies
$\in$ / $\notin$	$\in$ / $\notin$	BELONGS	Is / is not inside the group
$\cup$	$\cup$	UNION	Combine everything
$\cap$	$\cap$	INTERSECTION	Common elements
$-$	$-$	DIFFERENCE	In A but not in B
$\subseteq$ / $\subset$	$\subseteq$ / $\subset$	SUBSET	A is inside B / strict

LaTeX	Symbol	Concept	Meaning
			subset
<code>\$A^c\$</code>	A^c	COMPLEMENT	Everything outside A
<code>\$\emptyset\$</code>	$\emptyset$	EMPTY SET	A set with no elements
<code>\$U\$</code>	U	UNIVERSAL	The set of all possible elements

PART 1: LOGIC MASTER TRUTH TABLE

Instead of separate tables, use this to compare all logical operators at a glance:

A	B	$\neg A$ (NOT)	$A \wedge B$ (AND)	$A \vee B$ (OR)	$A \oplus B$ (XOR)	$A \rightarrow B$ (IMPLIES)	$A \leftrightarrow B$ (XNOR)
F	F	✓	F	F	F	✓	✓
F	✓	✓	F	✓	✓	✓	F
✓	F	F	F	✓	✓	F	F
✓	✓	F	✓	✓	F	✓	✓

De Morgan's Laws

How NOT interacts with AND / OR:

- $\neg(A \wedge B) \equiv \neg A \vee \neg B$
- $\neg(A \vee B) \equiv \neg A \wedge \neg B$

PART 2: QUANTIFIERS

Quantifiers describe how many elements satisfy a property.

FOR ALL ($\forall$)

Everything must satisfy the condition.

🧠 **Example:** $\forall x \in \mathbb{N} : x \geq 0$

EXISTS ($\exists$)

At least one satisfies the condition.

🧠 **Example:** $\exists x \in \mathbb{N} : x = 2$

UNIQUE EXISTENTIAL ($\exists!$) (New)

Exactly one element satisfies the condition.

🧠 **Example:** $\exists! x \in \mathbb{N} : x + 5 = 6$ (Only $x = 1$ works).

◆ PART 3: SET (GROUP) OPERATORS

Sets = groups of elements 📦

FOUNDATIONS

- **Empty Set ($\emptyset$):** The set containing nothing: $\{\}$.
- **Universal Set (U):** Every possible element in the current context.

BELONGING ($\in$ and $\notin$)

Is the element inside the group?

$A = \{1, 2, 3\}$

$2 \in A$ (True)

$5 \notin A$ (True)

UNION ($\cup$)

Combine everything + (NO duplicates)

$$A = \{1, 2\}, B = \{2, 3\}$$
$$A \cup B = \{1, 2, 3\}$$

INTERSECTION ($\cap$)

Common elements 🤝.

$$A = \{1, 2\}, B = \{2, 3\}$$
$$A \cap B = \{2\}$$

DIFFERENCE ($-$)

Elements in A but NOT in B.

$$A = \{1, 2, 3\}, B = \{2\}$$
$$A - B = \{1, 3\}$$

COMPLEMENT (A^c or $\neg A$)

Everything NOT in A (everything in U except A).

$$U = \{1, 2, 3, 4\}, A = \{1, 2\}$$
$$A^c = \{3, 4\}$$

SUBSET ($\subseteq$) & STRICT SUBSET ($\subset$)

- $A \subseteq B$: All of A is in B (they can be equal).
- $A \subset B$: All of A is in B , but B is strictly larger.

$A = \{1, 2\}$, $B = \{1, 2, 3\}$
 $A \subset B$ (True)



FINAL MEMORY TRICK

- **Logic** = truth values T (✓) or F
- **Sets** = groups 📦
- $\wedge$ = **strict**
- $\vee$ = **flexible**
- $\cap$ = **overlap**
- $\cup$ = **merge**